

F -purity of the maximal permanental ideal of a generic three-by-four matrix

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Abstract

Let X be a generic 3×4 matrix over a field of odd characteristic p , and let I be the ideal generated by its four maximal permanents. We prove that $k[X]/I$ is F -pure if and only if $p \equiv 1 \pmod{6}$, resolving a conjecture of Chau. The proof computes the coefficient that occurs in Fedder's criterion. Eliminating one row reduces this coefficient to a quartic determinant, whose constant-term description identifies it with a truncated Domb-number sum. A finite binomial transformation gives the exact value $\binom{(p-1)/2}{(p-1)/6}^2$ when $p \equiv 1 \pmod{6}$, and zero otherwise. We include an elementary complete-intersection argument in every odd characteristic and a telescoping proof of the classical finite transformation used in the calculation.

1 Introduction

For a matrix $X = (x_{ij})$ of independent variables, write $P_t(X)$ for the ideal generated by the permanents of its $t \times t$ submatrices. In odd characteristic these ideals behave differently from their determinantal counterparts. Even the ideals of maximal permanents of rectangular matrices present difficult questions about height, reducedness, and Frobenius singularities. Boralevi, Carlini, Michałek, and Ventura study their codimensions in [1]; Chau studies their F -singularities in [2].

Consider the generic matrix

$$X = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ v_1 & v_2 & v_3 & v_4 \end{pmatrix}.$$

Let f_j be the permanent of the 3×3 matrix obtained by deleting column j , and set $I = (f_1, f_2, f_3, f_4)$. Chau's Conjecture 4.5 in the preprint version of [2] predicts the following classification, which was checked there through characteristic 31.

Theorem 1.1. *Let k be any field of characteristic $p > 2$. For the generic 3×4 matrix X , the ring $k[X]/P_3(X)$ is F -pure if and only if $p \equiv 1 \pmod{6}$.*

Here F -purity means purity of the Frobenius homomorphism: tensoring the map $R \rightarrow F_*R$, $a \mapsto F_*(a^p)$, with any R -module preserves injectivity. The R -module structure on F_*R is $a \cdot F_*b = F_*(a^pb)$.

The central calculation is more precise than the nonvanishing statement needed for Theorem 1.1. Put

$$H = f_1 f_2 f_3 f_4, \quad M = \prod_{j=1}^4 a_j b_j v_j, \quad c(p) = [M^{p-1}] H^{p-1} \pmod{p}.$$

Brackets denote coefficient extraction in the indicated polynomial.

Theorem 1.2. *For every odd prime p ,*

$$c(p) = \begin{cases} \left(\frac{(p-1)/2}{(p-1)/6}\right)^2 & p \equiv 1 \pmod{6}, \\ 0 & p \not\equiv 1 \pmod{6} \end{cases} \quad \text{in } \mathbb{F}_p. \quad (1.1)$$

The proof identifies $c(p)$ with

$$\sum_{n=0}^{p-1} \frac{D_n}{4^n}, \quad D_n = \sum_{j=0}^n \binom{n}{j}^2 \binom{2j}{j} \binom{2n-2j}{n-j}. \quad (1.2)$$

The integers D_n are the Domb numbers. Their transformations and congruences are established number theory. In particular, Mao and Liu prove stronger congruences modulo p^3 in [4]. The contribution here is the reduction of the permanent coefficient to (1.2). We evaluate the sum modulo p using the finite transformation of Sun [5, Lemma 3.1], also recorded as equation (2.5) in the preprint version of [4]. Appendix A supplies a proof of that transformation, so no supercongruence is needed as an input.

All references to numbered statements of Chau and Mao–Liu use the specific preprint versions linked in the bibliography. The corresponding journal publication data are also supplied.

2 The complete-intersection and coefficient criteria

2.1 An elementary height computation

Let A consist of the first two rows of X . Expansion along the last row gives

$$(f_1, f_2, f_3, f_4)^t = L(A)v, \quad L_{ii} = 0, \quad L_{ij} = a_k b_l + a_l b_k \quad (i \neq j), \quad (2.1)$$

where $\{k, l\} = \{1, 2, 3, 4\} \setminus \{i, j\}$ and $v = (v_1, v_2, v_3, v_4)^t$. The matrix L is symmetric.

Proposition 2.1. *Over every field of characteristic different from two, the four polynomials f_1, f_2, f_3, f_4 form a regular sequence. In particular, I has height four.*

Proof. It suffices to compute dimensions over an algebraic closure. For $0 \leq r \leq 4$, let S_r be the locus in the eight-dimensional A -space on which L has rank r . We show

$$\dim S_r \leq 4 + r. \quad (2.2)$$

For $r = 4$ this is immediate. The determinant of L is a nonzero polynomial, as the specialization below shows, so the rank-at-most-three locus has codimension at least one.

The rank-at-most-two locus is contained in the common zero set of $\det L$ and the principal minor $2L_{12}L_{13}L_{23}$. These two polynomials have no common irreducible factor. Each L_{ij} is an irreducible polynomial of the form $xy + zw$ in four independent variables. To see that L_{12} does not divide $\det L$, specialize the four columns of A to

$$(1, 0)^t, \quad (0, 1)^t, \quad (1, 1)^t, \quad (1, -1)^t.$$

Then

$$L_{12} = 0, \quad (L_{13}, L_{14}, L_{23}, L_{24}, L_{34}) = (1, 1, -1, 1, 1),$$

and

$$\det L = (L_{13}L_{24} - L_{14}L_{23})^2 = 4 \neq 0.$$

Permuting columns gives the same conclusion for the other factors. Since height-one prime ideals in a polynomial unique factorization domain are principal, the indicated common zero set has codimension at least two. This proves (2.2) for $r = 2$.

A symmetric matrix with zero diagonal and rank at most one is zero: its principal 2×2 minors are $-L_{ij}^2$. It remains to bound the dimension of $L = 0$. These equations say that all 2×2 permanents of A vanish. If some column is $(a, b)^t$ with $ab \neq 0$, every other column has the form $t_j(a, -b)^t$. The permanent of two such columns is $-2abt_jt_k$; hence at most one of the other columns is nonzero. Each of these finitely many loci has dimension at most three. Otherwise every column is on a coordinate axis. Nonzero columns on opposite axes have nonzero permanent, so all nonzero columns must lie on the same axis. These loci have dimension four. Thus $\dim S_0 \leq 4$, and the rank-one stratum is empty.

Above S_r , equation (2.1) has a vector-space fiber of dimension $4 - r$. Consequently every rank stratum in $V(I)$ has dimension at most $(4 + r) + (4 - r) = 8$. The height theorem for four generators gives the reverse bound in the twelve-dimensional ambient space. Thus $\text{ht } I = 4$. A list of four homogeneous generators of height four in a polynomial ring is a regular sequence, since the ring is Cohen–Macaulay. $\square$

2.2 Fedder's criterion

For an ideal J in characteristic p , write $J^{[p]}$ for the ideal generated by p th powers of its elements. We use the following form of Fedder's criterion [3]: if S is a localization of a polynomial ring over $\mathbb{F}_p$ at a maximal ideal $\mathfrak{m}$, and $I \subseteq \mathfrak{m}$, then S/I is F -pure if and only if $(I^{[p]} : I) \not\subseteq \mathfrak{m}^{[p]}$. For a regular sequence $f_1, \dots, f_s$, the colon ideal is

$$(I^{[p]} : I) = I^{[p]} + ((f_1 \cdots f_s)^{p-1}). \quad (2.3)$$

We apply this first over $\mathbb{F}_p$ at the homogeneous maximal ideal.

Proposition 2.2. *For every field k of odd characteristic p , the ring $k[X]/I$ is F -pure if and only if $c(p) \neq 0$.*

Proof. The polynomial H^{p-1} has total degree $12(p-1)$. A monomial outside the ideal generated by the p th powers of the twelve variables must have every exponent at most $p-1$. Its degree therefore forces it to be M^{p-1} . Proposition 2.1 and (2.3) show that $c(p) = 0$ implies failure of F -purity at the origin over $\mathbb{F}_p$.

For sufficiency, there is a global splitting. Choose an additive map $\lambda : k \rightarrow k$ satisfying $\lambda(a^pb) = a\lambda(b)$ and $\lambda(1) = 1$. Such a map exists by choosing a k^p -basis of k containing 1, projecting to its 1-coordinate, and taking a p th root. On a monomial in the twelve variables define

$$T(bX^\alpha) = \begin{cases} \lambda(b)X^{(\alpha-(p-1)\mathbf{1})/p}, & \alpha_i \equiv p-1 \pmod{p} \text{ for every } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then $T(a^pg) = aT(g)$ and $T(H^{p-1}) = c(p)$. If $c(p) \neq 0$, the map $\phi(g) = c(p)^{-1}T(H^{p-1}g)$ satisfies $\phi(1) = 1$. Moreover,

$$H^{p-1}f_jg = f_j^p \prod_{i \neq j} f_i^{p-1}g,$$

so ϕ preserves every (f_j) and hence I . It induces a Frobenius splitting of $k[X]/I$, which implies F -purity.

Finally, F -purity descends along faithfully flat ring extensions. Thus F -purity over k would imply F -purity over $\mathbb{F}_p$, since all the equations are defined over $\mathbb{F}_p$. The necessity already proved over $\mathbb{F}_p$ therefore applies to every k . $\square$

3 Eliminating one row

We now work over $\mathbb{F}_p$. The elementary power sums are

$$\sum_{a \in \mathbb{F}_p} a^e = \begin{cases} -1, & e > 0 \text{ and } p-1 \mid e, \\ 0, & \text{otherwise,} \end{cases}$$

where the case $e = 0$ is the sum of p copies of 1. Consequently, for a homogeneous polynomial G of degree $N(p-1)$,

$$\sum_{x \in \mathbb{F}_p^N} G(x) = (-1)^N [x_1^{p-1} \cdots x_N^{p-1}] G. \quad (3.1)$$

Indeed, a monomial can survive only if every exponent is a positive multiple of $p-1$, and total degree then forces equality in every coordinate.

Proposition 3.1. *With L as in (2.1),*

$$c(p) = \left[\prod_{j=1}^4 a_j^{p-1} b_j^{p-1} \right] \det L(A)^{p-1}. \quad (3.2)$$

Proof. Apply (3.1) in twelve variables and sum over v first. If $L(A)$ is singular, each nonempty fiber of the map $v \mapsto L(A)v$ has cardinality divisible by p . The sum of $\prod_j (L(A)v)_j^{p-1}$ over all v is therefore zero in $\mathbb{F}_p$. If $L(A)$ is invertible, a change of variables gives

$$\sum_{v \in \mathbb{F}_p^4} \prod_{j=1}^4 (L(A)v)_j^{p-1} = \prod_{j=1}^4 \sum_{w \in \mathbb{F}_p} w^{p-1} = 1.$$

In either case the sum is $\det L(A)^{p-1}$. Applying (3.1) in the remaining eight variables proves (3.2). $\square$

3.1 The quartic determinant

Make the formal Laurent-polynomial substitution $a_i = b_i t_i$ and write $e_j = e_j(t_1, t_2, t_3, t_4)$ for the elementary symmetric polynomials. Since each variable column has degree two in $\det L$,

$$\det L(A) = \left(\prod_{i=1}^4 b_i^2 \right) D(t), \quad D(t) = -4(e_1 e_3 - 4e_4). \quad (3.3)$$

Here is an explicit verification. For a symmetric matrix with zero diagonal and upper-triangular entries (a, b, c, d, e, f) in row order, the determinant is

$$a^2 f^2 + b^2 e^2 + c^2 d^2 - 2(abef + acdf + bcde).$$

After removing the b_i factors from L , the six entries are

$$t_3 + t_4, t_2 + t_4, t_2 + t_3, t_1 + t_4, t_1 + t_3, t_1 + t_2.$$

Substitution in the displayed determinant gives $-4(e_1e_3 - 4e_4)$. Equivalently, its twelve monomials of exponent pattern $(2, 1, 1, 0)$ have coefficient -4 , and all other coefficients are zero.

If t_i has exponent α_i in $D(t)^{p-1}$, then after substitution the exponents on a_i, b_i are respectively $\alpha_i, 2(p-1) - \alpha_i$. Thus

$$c(p) = [(t_1t_2t_3t_4)^{p-1}]D(t)^{p-1}. \quad (3.4)$$

This is a polynomial coefficient calculation; it does not discard points at which some b_i vanishes.

Set

$$Q(t) = \left(\sum_{i=1}^4 t_i \right) \left(\sum_{i=1}^4 t_i^{-1} \right).$$

Since $e_1e_3 = e_4Q$ and $(-4)^{p-1} = 1$ in $\mathbb{F}_p$, (3.4) becomes

$$c(p) = \text{CT}(Q - 4)^{p-1} = \sum_{n=0}^{p-1} 4^{-n} \text{CT } Q^n. \quad (3.5)$$

The second equality follows from $\binom{p-1}{n} \equiv (-1)^n \pmod{p}$.

Lemma 3.2. *For every nonnegative integer n , $\text{CT } Q^n = D_n$.*

Proof. The multinomial theorem gives

$$\text{CT } Q^n = \sum_{n_1+n_2+n_3+n_4=n} \binom{n}{n_1, n_2, n_3, n_4}^2.$$

Group terms according to $j = n_1 + n_2$, and use $\sum_{a=0}^j \binom{j}{a}^2 = \binom{2j}{j}$ for each of the two pairs. The resulting sum is precisely the definition of D_n in (1.2). $\square$

4 Evaluation of the Domb-number sum

We use the integer identity

$$D_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n+k}{3k} \binom{2k}{k}^2 \binom{3k}{k} 4^{n-2k}. \quad (4.1)$$

It is a classical transformation; see [5, Lemma 3.1] and [4, equation (2.5)]. A complete telescoping proof is given in Appendix A.

Proposition 4.1. *For an odd prime p , let $C_p = \sum_{n=0}^{p-1} D_n/4^n \in \mathbb{F}_p$. If $p \not\equiv 1 \pmod{3}$, then $C_p = 0$. If $p \equiv 1 \pmod{3}$ and $m = (p-1)/3$, then*

$$C_p = 16^{-m} \binom{2m}{m}^2 \neq 0. \quad (4.2)$$

Proof. Insert (4.1), interchange the finite sums, and use the elementary summation identity $\sum_{n=2k}^{p-1} \binom{n+k}{3k} = \binom{p+k}{3k+1}$. This gives, first as an equality of rational numbers and then modulo p ,

$$C_p = \sum_{k=0}^{(p-1)/2} 16^{-k} \binom{2k}{k}^2 \binom{3k}{k} \binom{p+k}{3k+1}. \quad (4.3)$$

The product of the last two binomial coefficients is

$$\binom{3k}{k} \binom{p+k}{3k+1} = \frac{\prod_{j=-2k}^k (p+j)}{k!(2k)!(3k+1)}. \quad (4.4)$$

Because $0 \leq k \leq (p-1)/2$, the numerator has exactly one factor divisible by p , namely the factor with $j = 0$; it is exactly p . The factors $k!$ and $(2k)!$ are units modulo p . Also $1 \leq 3k+1 < 2p$, so the only possible cancellation of that factor is $3k+1 = p$.

If $p \not\equiv 1 \pmod{3}$, there is no such k and every summand vanishes. This includes $p = 3$. Otherwise put $m = (p-1)/3$. At $k = m$ we cancel p in (4.4) over the integers before reducing; its value modulo p is

$$\frac{(-1)^{2m} (2m)! m!}{m! (2m)!} = 1.$$

Only this summand remains, proving (4.2). The binomial coefficient is nonzero modulo p because $2m < p$. $\square$

Proof of Theorems 1.1 and 1.2. Equation (3.5) and Lemma 3.2 give $c(p) = C_p$. For odd p , the condition $p \equiv 1 \pmod{3}$ is equivalent to $p \equiv 1 \pmod{6}$. Proposition 2.2 and Proposition 4.1 prove Theorem 1.1.

For the exact coefficient, use the rational identity

$$\binom{2m}{m} = (-4)^m \binom{-1/2}{m}.$$

When $m = (p-1)/3$, all denominators are invertible modulo p . Reducing $-1/2$ to $(p-1)/2$ and using binomial symmetry gives

$$16^{-m} \binom{2m}{m}^2 = \binom{(p-1)/2}{m}^2 = \binom{(p-1)/2}{(p-1)/6}^2 \quad \text{in } \mathbb{F}_p.$$

This proves Theorem 1.2. $\square$

5 Consequences and comparison

When $p \equiv 1 \pmod{6}$, the proof supplies an explicit compatible Frobenius splitting, and hence reducedness of the maximal permanental quotient. In every odd characteristic, independently of F -purity, Proposition 2.1 gives the Hilbert series

$$\text{Hilb}_{k[X]/I}(z) = \frac{(1-z^3)^4}{(1-z)^{12}}.$$

Failure of F -purity in the other characteristics is not, by itself, a statement that the quotient is nonreduced.

The residue calculation is consistent with the stronger arithmetic result of Mao–Liu [4, Theorem 1.1]. For $p > 3$ that theorem implies, modulo p , that the sum in (1.2) equals $4x^2$ if $p = x^2 + 3y^2 \equiv 1 \pmod{3}$, and zero if $p \equiv 2 \pmod{3}$. Thus it provides a second way to conclude the nonvanishing criterion once the permanental reduction is established. The proof above uses only the finite transformation and does not assert a new congruence for the Domb numbers.

For illustration, the first few coefficient values are

p	3	5	7	11	13	17	19	23
$c(p)$	0	0	2	0	4	0	7	0.

Exact finite checks of the determinant identity and the telescoping identities accompany the manuscript. The classification for all odd primes follows from the proofs, rather than from extrapolation of these values.

A A telescoping proof of the finite transformation

This appendix proves the established identity (4.1) over the integers. The proof is included to make the arithmetic input directly checkable.

Put

$$\begin{aligned} f(n, k) &= \binom{n}{k}^2 \binom{2k}{k} \binom{2n-2k}{n-k}, & 0 \leq k \leq n, \\ g(n, k) &= \binom{n+k}{3k} \binom{2k}{k}^2 \binom{3k}{k} 4^{n-2k}, & 0 \leq 2k \leq n, \end{aligned}$$

and extend each summand by zero outside its stated range. For $n \geq 2$ define

$$\begin{aligned} B_n &= 2(2n-1)(5n(n-1)+2), \\ P(n, k) &= 4k^3 + (6-18n)k^2 + (26n^2-15n)k + 10n^2 - 12n^3, \\ r(n, k) &= \frac{k^3 P(n, k)}{n^2(2n-2k-1)}, \\ s(n, k) &= \frac{-12(3n-2)k^4}{(n+k)(n+k-1)}. \end{aligned}$$

The two telescoping identities are

$$\begin{aligned} n^3 f(n, k) - B_n f(n-1, k) + 64(n-1)^3 f(n-2, k) \\ = r(n, k+1) f(n, k+1) - r(n, k) f(n, k), \end{aligned} \tag{A.1}$$

$$\begin{aligned} n^3 g(n, k) - B_n g(n-1, k) + 64(n-1)^3 g(n-2, k) \\ = s(n, k+1) g(n, k+1) - s(n, k) g(n, k). \end{aligned} \tag{A.2}$$

We give polynomial forms of these identities to avoid an implicit appeal to symbolic summation software. In (A.1) put $h = n - k$. Cancellation of factorials and clearing denominators reduces it to

$$\begin{aligned} 2(2k+1)h^3 P(n, k+1) - 2k^3(2h-3)P(n, k) \\ = 2n^5(2h-1)(2h-3) - B_n h^3(2h-3) + 32(n-1)h^3(h-1)^3. \end{aligned} \tag{A.3}$$

In (A.2) put $h = n - 2k$ and $C = -12(3n - 2)$. The analogous polynomial identity is

$$\begin{aligned} C((n+k-1)(2k+1)h(h-1) - 8k^4) \\ = 8n^3(n+k)(n+k-1) - 2B_n h(n+k-1) + 32(n-1)^3 h(h-1). \end{aligned} \quad (\text{A.4})$$

Both (A.3) and (A.4) follow by expansion in $\mathbb{Z}[n, k]$. To recover the summand identities, the required successive ratios are

$$\begin{aligned} \frac{f(n, k+1)}{f(n, k)} &= \frac{(2k+1)(n-k)^3}{(k+1)^3(2n-2k-1)}, \\ \frac{f(n-1, k)}{f(n, k)} &= \frac{(n-k)^3}{2n^2(2n-2k-1)}, \\ \frac{f(n-2, k)}{f(n, k)} &= \frac{(n-k)^3(n-k-1)^3}{4n^2(n-1)^2(2n-2k-1)(2n-2k-3)}, \\ \frac{g(n, k+1)}{g(n, k)} &= \frac{(n+k+1)(2k+1)(n-2k)(n-2k-1)}{8(k+1)^4}, \\ \frac{g(n-1, k)}{g(n, k)} &= \frac{n-2k}{4(n+k)}, \\ \frac{g(n-2, k)}{g(n, k)} &= \frac{(n-2k)(n-2k-1)}{16(n+k)(n+k-1)}. \end{aligned}$$

The factors in the numerators account for zero summands at the upper boundary. None of the displayed denominators in the f ratios vanishes on its integer summation range when $n \geq 2$; the same is true of the g ratios. The certificates are finite at the endpoints, and $r(n, 0) = s(n, 0) = 0$. At the upper endpoint plus one, the extended summand is zero. Summing (A.1) or (A.2) therefore gives the recurrence

$$n^3 U_n = B_n U_{n-1} - 64(n-1)^3 U_{n-2} \quad (n \geq 2).$$

Both finite sums have $U_0 = 1$ and $U_1 = 4$. Over $\mathbb{Q}$, these initial values and the nonzero coefficient n^3 determine the sequence uniquely. The sums agree as integers for every n , proving (4.1). Reduction modulo a prime is performed only after this integer identity has been established.

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